

datasheet

ESP32 S2 Mini

ESP32-S2FN4R2

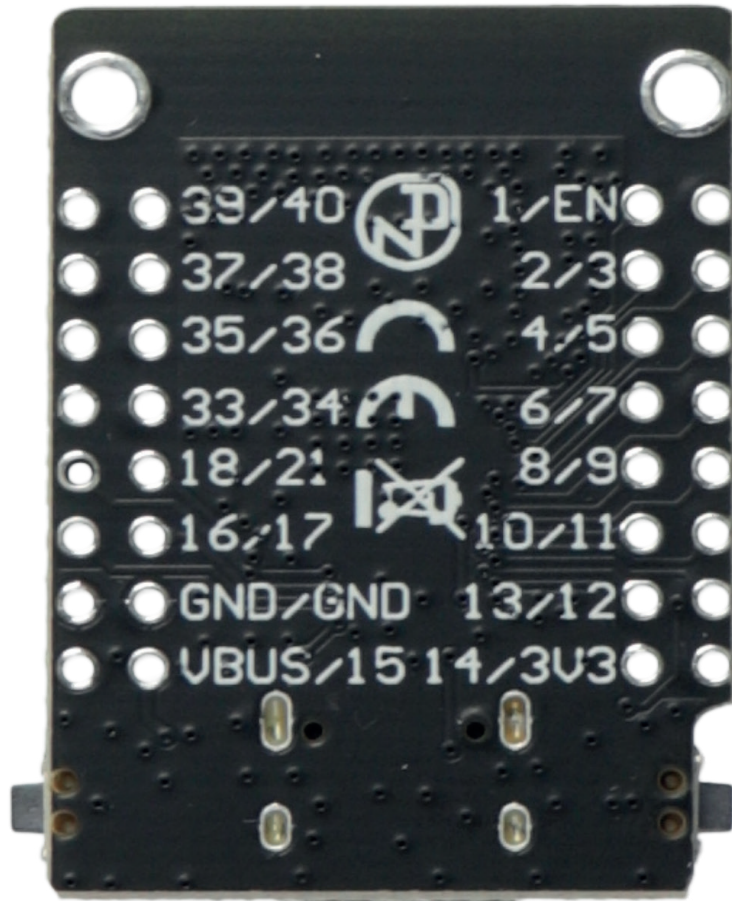


Discover the capabilities of our ESP32-S2FN4R2-based board, featuring a Type-C USB interface, 4MB Flash, 2MB PSRAM, and 27x IO options. With built-in ADC, DAC, I2C, SPI, UART, and USB OTG functionalities, this platform offers versatility for various projects. It supports MicroPython, Arduino, CircuitPython, and ESP-IDF, providing flexibility for your development needs.

Specifications

MCU	ESP32-S2FN4R2
Operating Voltage & logic Voltage	3.3V
Flash size	4MB
Embedded PSRAM	2MB
Clock Speed	240MHz
Interfaces	SPI, I2C, I2S, UART, DAC, ADC, USB-OTG
GPIOs	27
Dimensions	25 x 34.5 x 4.2 mm
Weight	2.4g

Pin definition



Pin	Description
1	RTC_GPIO1, GPIO1, TOUCH1, ADC1_CH0
2	RTC_GPIO2, GPIO2, TOUCH2, ADC1_CH1
3	RTC_GPIO3, GPIO3, TOUCH3, ADC1_CH2
4	RTC_GPIO4, GPIO4, TOUCH4, ADC1_CH3
5	RTC_GPIO5, GPIO5, TOUCH5, ADC1_CH4
6	RTC_GPIO6, GPIO6, TOUCH6, ADC1_CH5

7	RTC_GPIO7, GPIO7, TOUCH7, ADC1_CH6
8	RTC_GPIO8, GPIO8, TOUCH8, ADC1_CH7
9	RTC_GPIO9, GPIO9, TOUCH9, ADC1_CH8, FSPIHD
10	RTC_GPIO10, GPIO10, TOUCH10, ADC1_CH9, FSPICS0, FSPIIO4
11	RTC_GPIO11, GPIO11, TOUCH11, ADC2_CH0, FSPID, FSPIIO5
12	RTC_GPIO12, GPIO12, TOUCH12, ADC2_CH1, FSPICLK, FSPIIO6
13	RTC_GPIO13, GPIO13, TOUCH13, ADC2_CH2, FSPIQ, FSPIIO7
14	RTC_GPIO14, GPIO14, TOUCH14, ADC2_CH3, FSPIWP, FSPIDQS
15	RTC_GPIO15, GPIO15, U0RTS, ADC2_CH4, XTAL_32K_P
16	RTC_GPIO16, GPIO16, U0CTS, ADC2_CH5, XTAL_32K_N
17	RTC_GPIO17, GPIO17, U1TXD, ADC2_CH6, DAC_1
18	RTC_GPIO18, GPIO18, U1RXD, ADC2_CH7, DAC_2, CLK_OUT3
21	RTC_GPIO21, GPIO21
33	SPIIO4, GPIO33, FSPIHD
34	SPIIO5, GPIO34, FSPICS0
35	SPIIO6, GPIO35, FSPID
36	SPIIO7, GPIO36, FSPICLK
37	SPIDQS, GPIO37, FSPIQ
38	GPIO38, FSPIWP
39	MTCK, GPIO39, CLK_OUT3
40	MTDO, GPIO40, CLK_OUT2
EN	CHIP_PU, High: on, enables the chip. Low: off, the chip powers off
3.3V	3.3V pin
GND	Ground

VBUS	VBUS output or 5V input pin
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For further information about the ESP32-S2FN4R2 mcu please check out
Espressif datasheet at:

https://www.espressif.com/sites/default/files/documentation/esp32-s2_datasheet_en.pdf